

Do LLM Rankings Change if You Ask Differently?

Ranking reversals under non-semantic prompt perturbations
and subset resampling

ST5230 Group 14

A Familiar Story, and the Question Behind It

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- Model A: 89.2%
- Model B: 87.3%
- Therefore, A is better than B

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Our target is not only score accuracy, but the reliability of the ranking itself.

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- In practice, benchmark outcomes depend on at least two nontrivial factors:
 - prompt formulation
 - evaluation subset composition
- If these sources of variation are large relative to model gaps, rank claims may be overconfident.

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Benchmark papers

MMLU, ARC, and HellaSwag establish the task settings we evaluate.

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Our contribution

We focus on **ranking stability**: not just whether scores move, but whether the order between models reverses.

Research Questions

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If the answer is no

A static leaderboard is a reasonable summary.

If the answer is yes

Rank claims should be reported with uncertainty and robustness diagnostics.

Method

Treat ranking as a random outcome induced by evaluation choices

Core Idea and Theoretical Framing

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Interpretation

Rank is not treated as a fixed truth, but as a distribution over plausible evaluation choices.

High RRR means a single leaderboard is a fragile snapshot rather than a stable conclusion.

Experimental Setup

Benchmarks

- MMLU: 300 questions
- ARC-Challenge: 300 questions
- HellaSwag: 300 questions

Prompt templates

- `minimal_instruction`
- `benchmark_style`
- `natural_style`
- `order_phrasing_variation`

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- GPT-4o-mini
- Gemini-2.0-flash
- Qwen-2.5-7B
- Llama-3.1-8B

Inference / analysis

- `temperature = 0`
- `max_tokens = 150`
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900

questions

3600

prompt requests

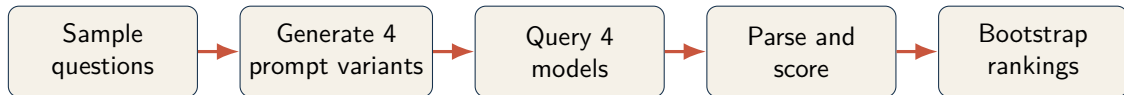
14400

responses

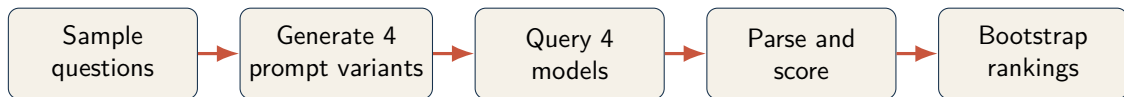
3

parse failures

Pipeline



Pipeline

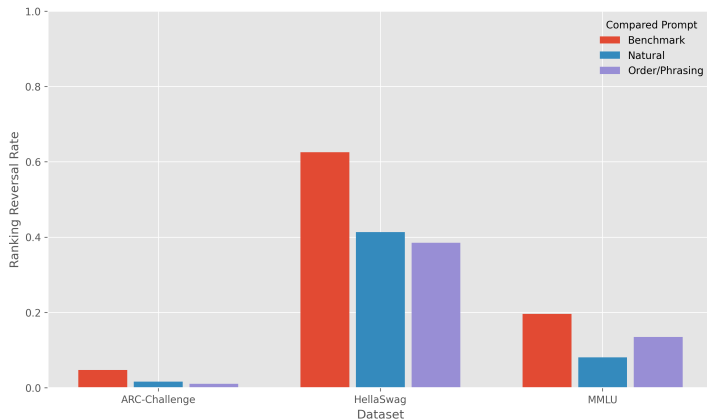


- This design isolates two realistic sources of instability: **prompt choice** and **subset choice**.
- It also keeps the task fixed, so reversals cannot be dismissed as semantic task drift.

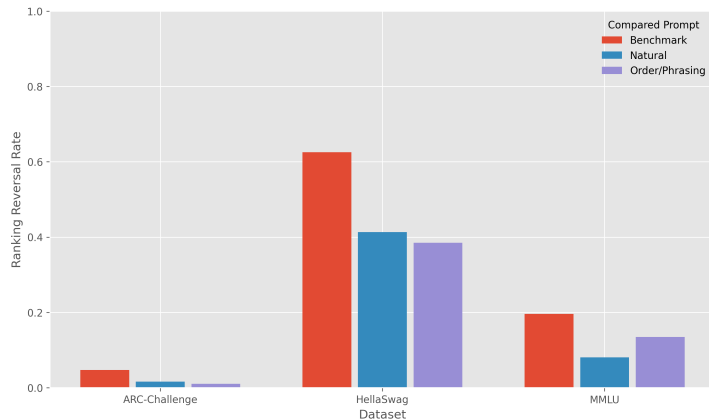
Results

Three benchmarks, three very different stability stories

Main Result: Ranking Stability Is Benchmark-dependent



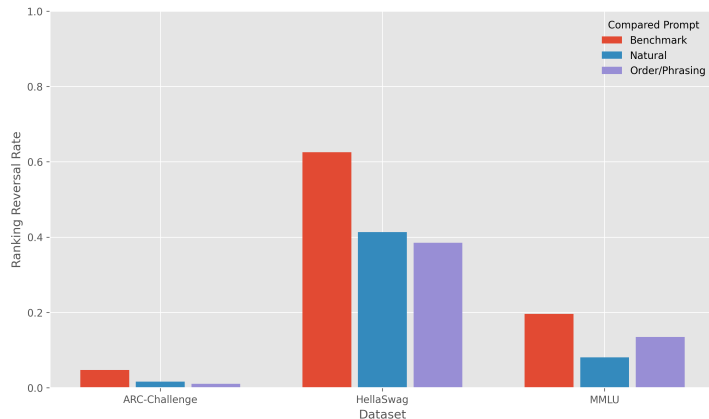
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RRR relative to the minimal prompt

- ARC: 4.70%, 1.65%, 1.05%
- MMLU: 19.60%, 8.05%, 13.50%

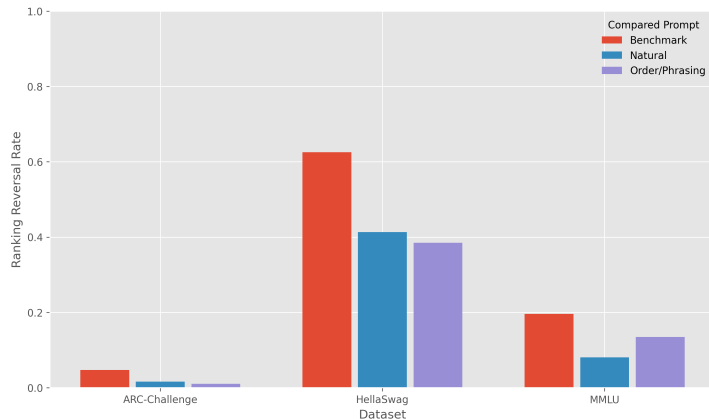
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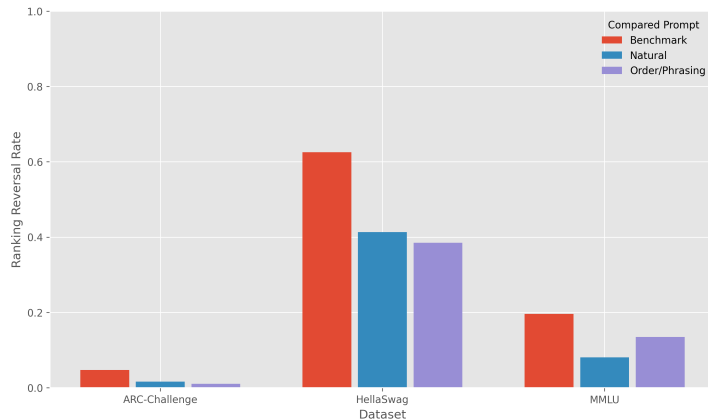
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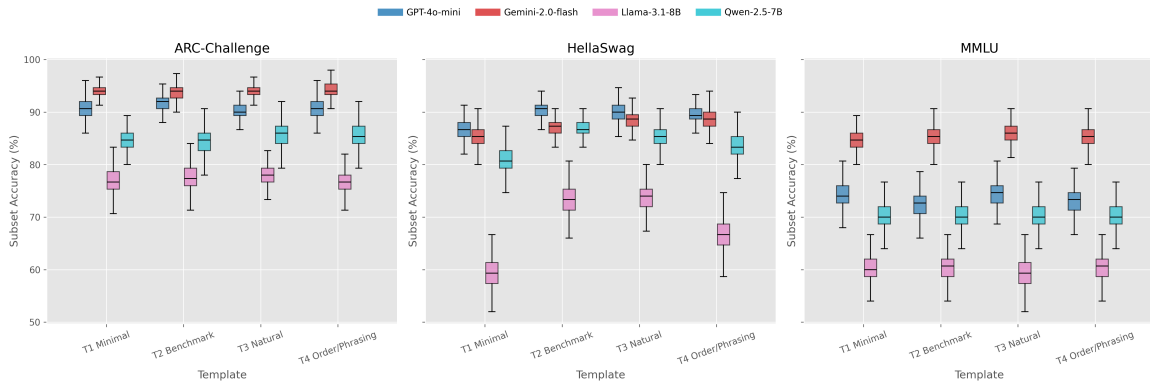


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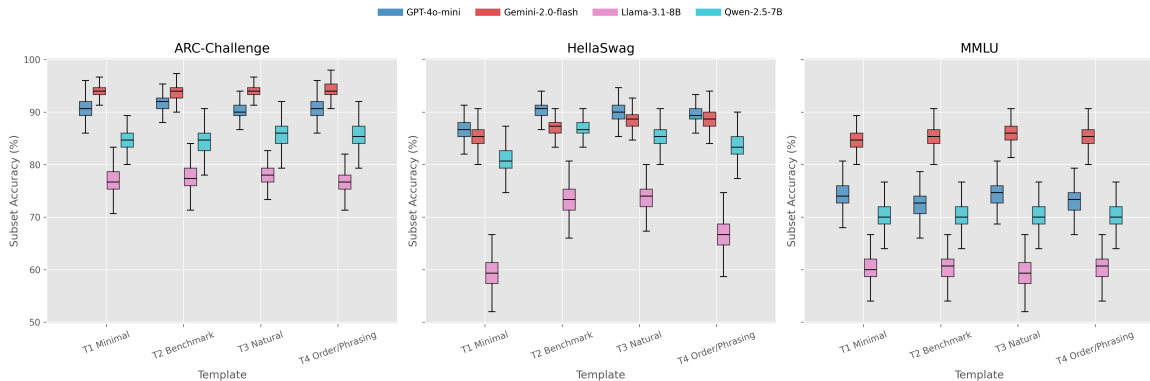
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Same method, same models,
same prompt philosophy;
radically different stability.

Reading the Accuracy Landscape

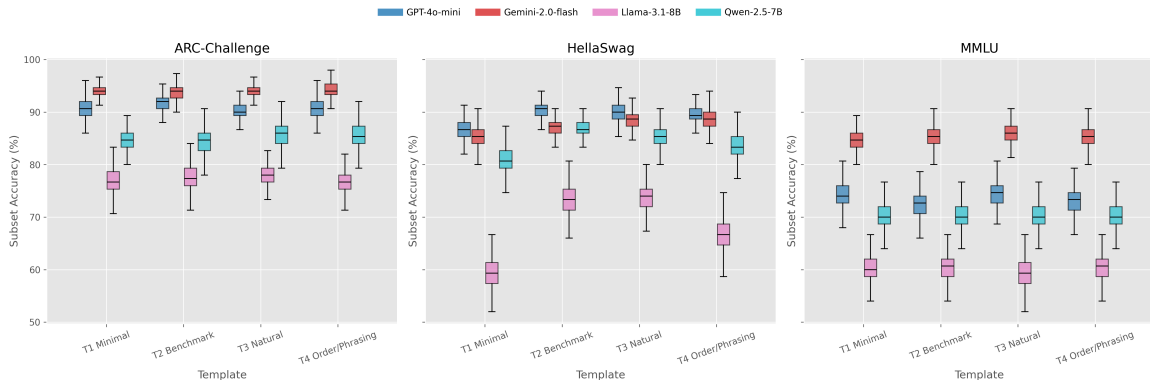


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- **ARC**: tight boxes, clear separation, wide performance margins.
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- **HellaSwag**: larger prompt sensitivity, especially around the top two models.
- **MMLU**: clear winner, but less separation in the middle ranks.

Case Study 1: HellaSwag and the Fragile Winner

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highest RRR under benchmark-style
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Interpretation

HellaSwag provides the clearest counterexample to the static leaderboard narrative: the apparent winner exists at the point-estimate level, but not as a robust rank claim.

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- Gemini is first under all prompt variants.

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ARC behaves like the idealized leaderboard story: prompt and subset variation are too weak to overturn the ordering.

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Gemini prompt-averaged accuracy

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Gemini top-1 probability across prompts

1–5%

RRR range only

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- So MMLU is not fully unstable; it is **hierarchically unstable**: the winner is settled, the middle ranks are not.

The right lesson is not “MMLU is noisy,” but “different parts of the ranking have different certainty levels.”

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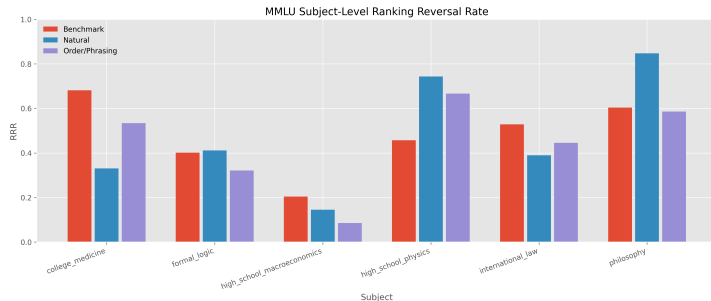
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GPT lead over Qwen

$p = 0.184$

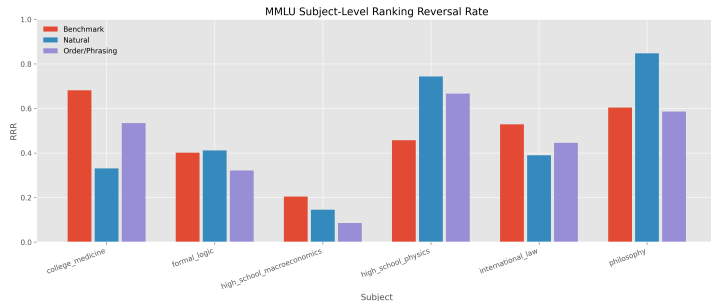
GPT–Qwen gap is not significant

One More Layer: MMLU Is Also Subject-dependent



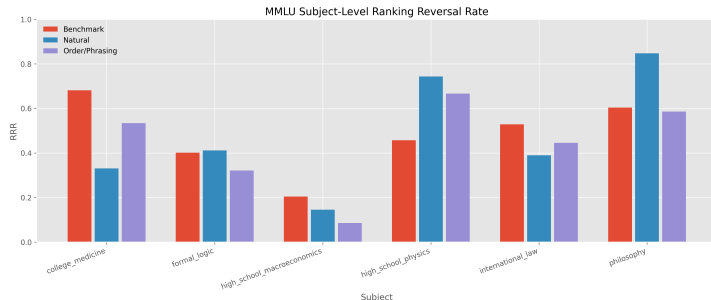
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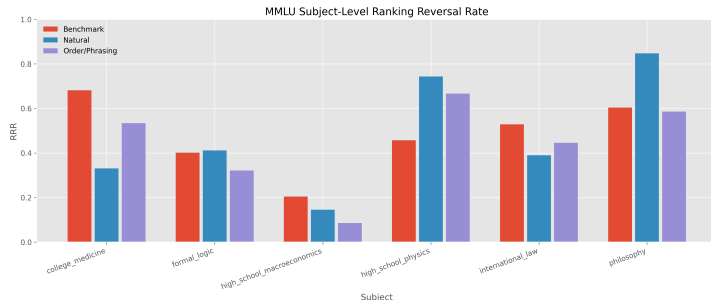
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Ranking robustness is benchmark-dependent **and** subdomain-dependent.

Takeaways

What this project says about how LLM evaluation should be reported

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A better benchmark report should state not only who is best on average, but how robust that ordering is.

Implications and Limitations

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- report confidence intervals and pairwise significance
- report ranking stability metrics such as RRR
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This is not a claim that all leaderboards are unreliable. It is a claim that reliability must be demonstrated, not assumed.

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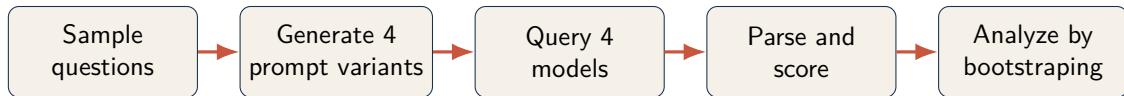
fragile top rank

The central question is not only “Who is best?” but “Does that ordering survive reasonable evaluation variation?”

Experiment Design and Result

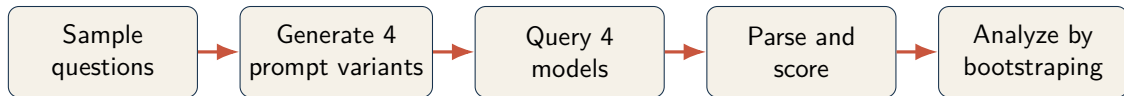
Experimental workflow, metric definitions, and reversal cases we found

Pipeline



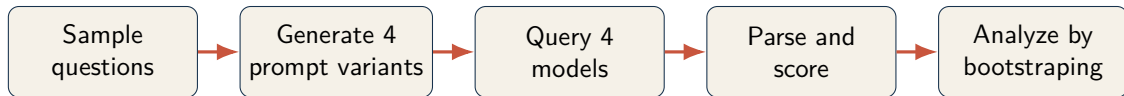
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- Phase 3: By bootstrapping, compute model accuracy and ranking reversal rates (see next slide).

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- **Prompt-wise Ranking Reversal Rate (PRRR)**: For an item set P , two templates T_1 and T_2 , draw N subsets $P_i \subset P$; the PRRR on P between T_1 and T_2 is the frequency of different rankings occurring under two templates among the N subsets, denote $\text{PRRR}(P, T_1, T_2)$. i.e.,

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- **Subset Ranking Reversal Rate (SRRR)**: For an item set P and template T , draw N subsets $P_i \subset P$; the SRRR on P under T is the frequency of different rankings occurring under T when asked subsets and the full set, denote $\text{SRRR}(P, T)$. i.e.,

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Definition of Core Statistics: Accuracy and RRR's

**We isolate two sources of instability: prompt choice and incomplete test set.
PRRR measures prompt-to-prompt rank disagreement on the same task set;
SRRR measures full-set vs subset rank instability under the same prompt.**

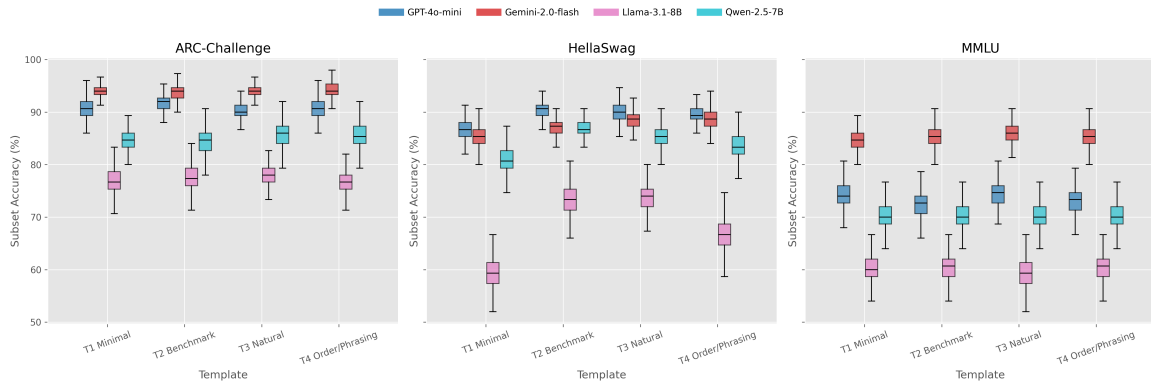
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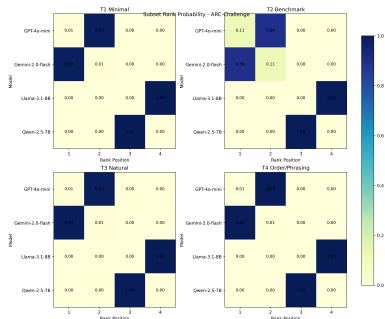
- We applied bootstrap number $N = 1000$, and drew half-sized subsets: $\#P_i = \#P/2$.
- For 3 datasets (MMLU, ARC-Challenge, and HellaSwag), and 6 subjects of question sets from MMLU, we computed:
 - PRRR between minimal instruction and the other 3 templates;
 - SRRR under all 4 templates.

Experiment Results: Dataset-Level Ranking Instability

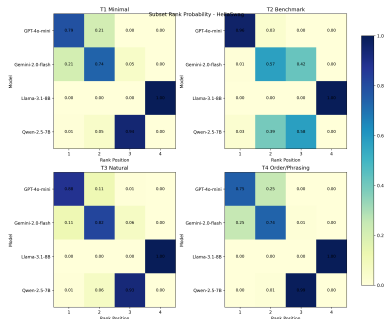


The accuracy boxplots show that Gemini leads on ARC-Challenge and MMLU, while GPT-4o-mini leads on HellaSwag.

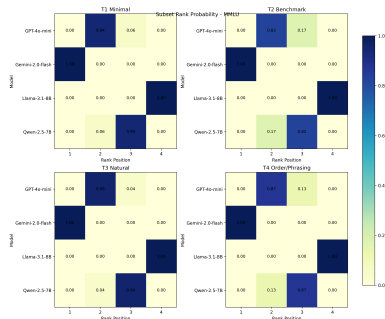
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Ranking heatmap (ARC)



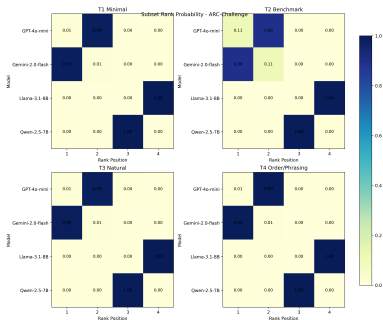
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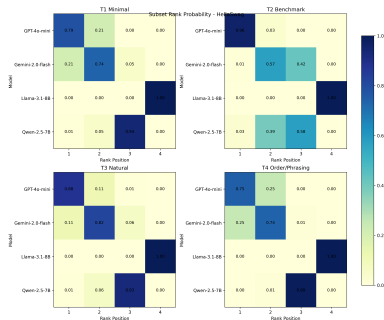
Ranking heatmap (MMLU)

- The darker a tile, the more likely a model falls into that place in a ranking.

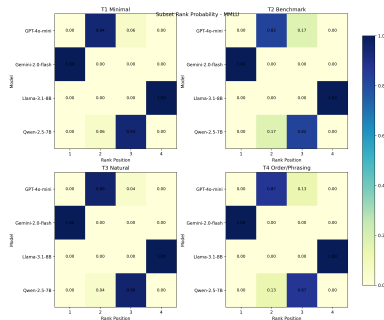
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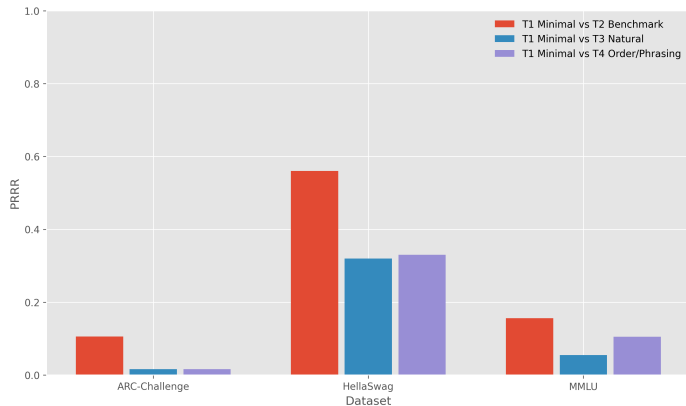
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Ranking heatmap (MMLU)

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- Distribution of model rankings differ among datasets and prompt styles.

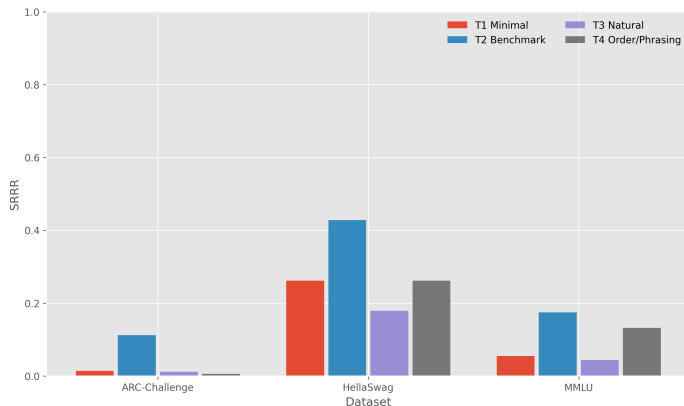
Experiment Results: Dataset-Level RRR's



PRRR Results:

- ARC-Challenge:
max PRRR = 10.6%;
- HellaSwag:
max PRRR = 56.0%;
- MMLU:
max PRRR = 15.6%.

Experiment Results: Dataset-Level RRR's



SRRR Results:

- ARC-Challenge:
max SRRR = 11.2%
- HellaSwag:
max SRRR = 42.8%
- MMLU:
max SRRR = 17.5%

Experiment Results: Dataset-Level RRR's

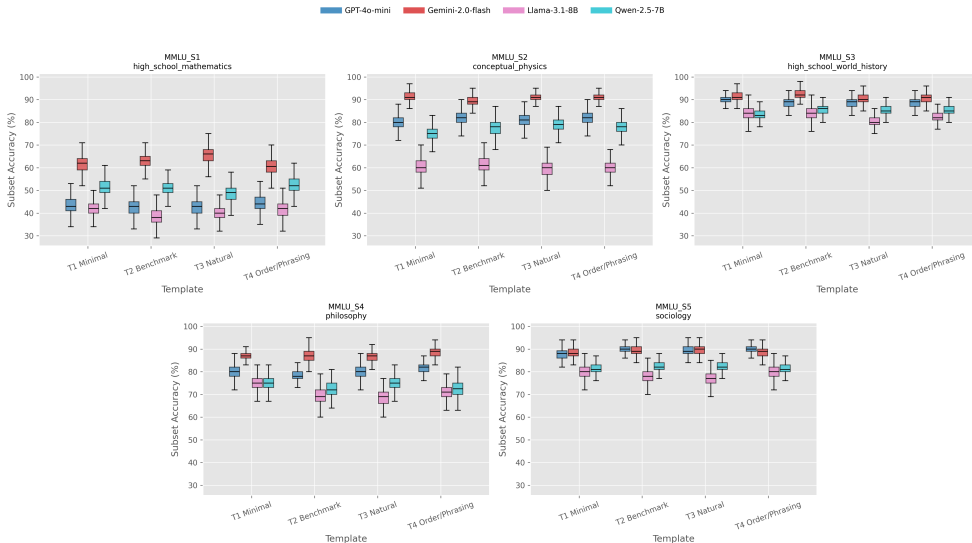
- **HellaSwag**: strongest dataset-level reversal case, especially under T2 Benchmark.
- **MMLU**: overall winner is stable, but middle ranks are more sensitive to subset changes.
- **ARC**: rank order survives both prompt and subset variation in most resamples.

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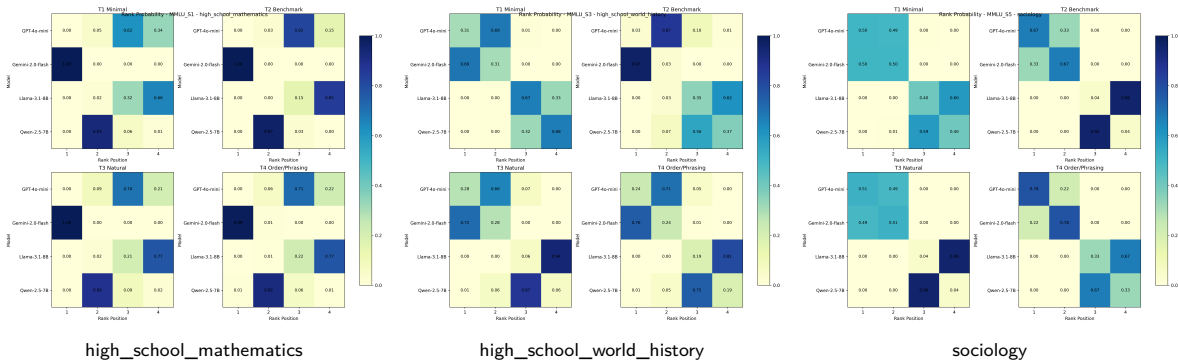
- **HellaSwag**: strongest dataset-level reversal case, especially under T2 Benchmark.
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HellaSwag is the clearest fragile-leaderboard case. ARC-Challenge remains the most stable benchmark.

Experiment Results: MMLU Subject-Level Ranking Instability



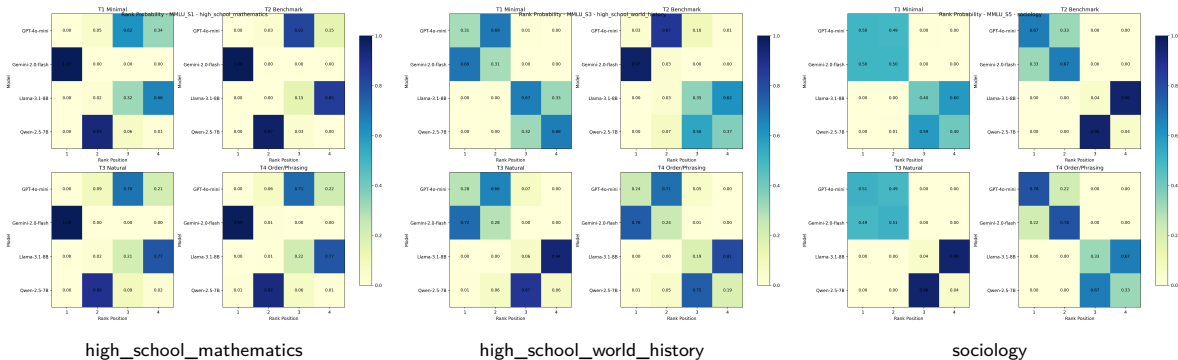
Experiment Results: MMLU Subject-Level Ranking Instability



Heatmaps show the rankings vary a lot among subjects and prompt styles.

- Model ranking distributions are very different among all 3 example subjects.

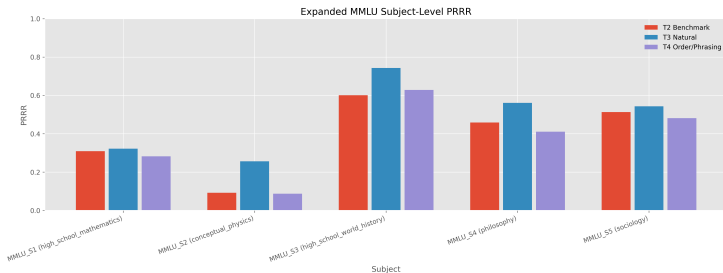
Experiment Results: MMLU Subject-Level Ranking Instability



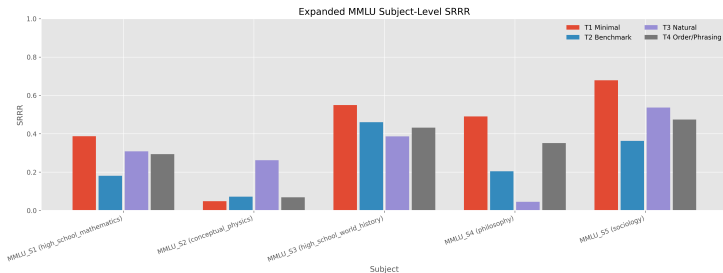
Heatmaps show the rankings vary a lot among subjects and prompt styles.

- Model ranking distributions are very different among all 3 example subjects.
- Different prompts can break a 50-50 tie between two models.

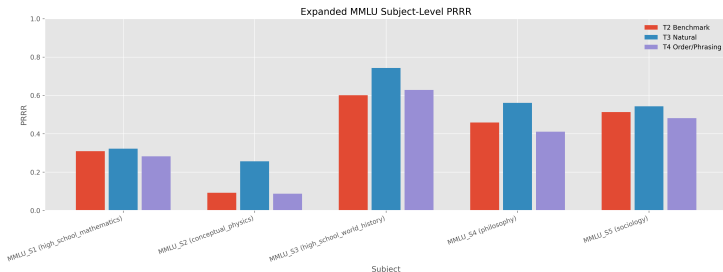
Experiment Results: MMLU Subject-Level RRR's



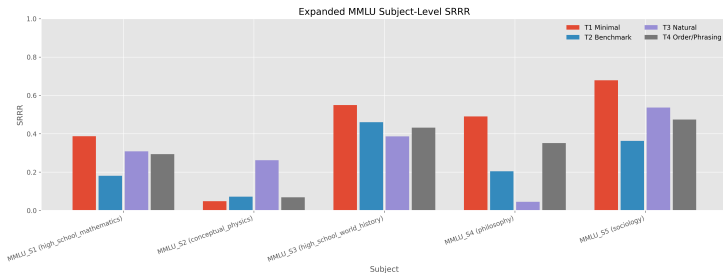
- Largest PRRR:
high_school_world_history,
T1 vs T3 = 74.30%



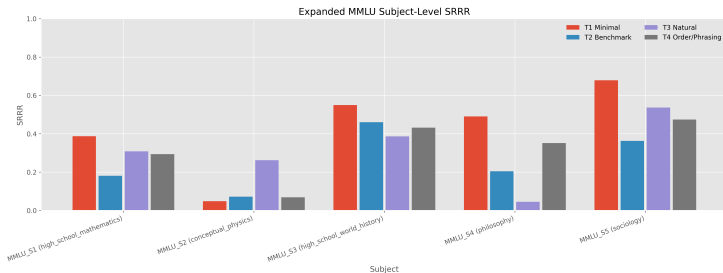
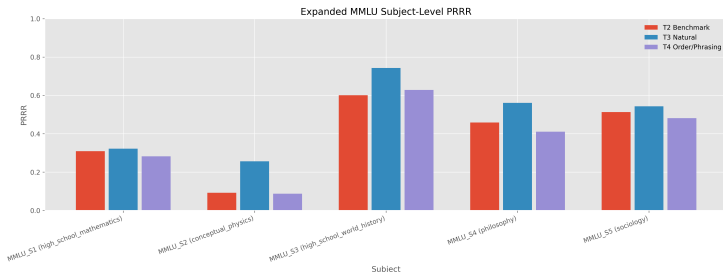
Experiment Results: MMLU Subject-Level RRR's



- Largest PRRR: high_school_world_history, T1 vs T3 = 74.30%
- Largest SRRR: sociology, T1 = 67.90%



Experiment Results: MMLU Subject-Level RRR's



- Largest PRRR: `high_school_world_history`, T1 vs T3 = 74.30%
- Largest SRRR: `sociology`, T1 = 67.90%

The RRR's show that subject-level ranking instability is even higher than dataset-Level instability.